

What This Data Can and Cannot Tell Us

Nipul Jayasekera (njayase1) · EN.605.256 Modern Software Concepts in Python · Module 3

Grad Cafe is not a sample of graduate applicants; it is a sample of graduate applicants who chose to post. Nothing about the site recruits participants, verifies them, or follows up with the ones who never returned, so every figure in this analysis describes the population of people who decided their outcome was worth typing into a web form. That population is assembled by self-selection at two separate stages, and both push in the same direction. An applicant must first decide to post at all — a decision that anecdotally correlates with having news worth sharing, whether triumphant or crushing, and with belonging to the online communities where the site circulates, which skews heavily toward computer science, engineering, the sciences and North American research universities. They must then decide, field by field, how much to disclose. The second decision is visible directly in my own data: 29,483 entries report a GPA but only 3,769 report any GRE Quantitative score, so the GRE averages in Question 3 rest on under eight per cent of the table. Missingness of that magnitude is not noise that averages out. If applicants who scored poorly are likelier to leave the box empty — which is the ordinary human expectation, and which nothing in the data can rule out — then the reported mean is an estimate of the mean among people willing to publish their score, and is biased upward as an estimate of anything wider. The same logic applies to the acceptance rate in Question 5: an applicant who is rejected everywhere has less reason to come back and update a profile than one with an offer to celebrate, so a 40.74% acceptance rate cannot be read as the probability that a given application succeeds.

The second limitation is that the data is not merely incomplete but partly wrong, in ways arithmetic will happily propagate. Grad Cafe validates nothing an applicant types, and the supporting analysis under Question 3 counts the consequences: 2,316 of the 3,769 reported GRE Quantitative values — 61 per cent of them — are impossible on a scale that runs 130 to 170. They are not random noise. The GRE reports Verbal and Quantitative on 130-170 each and a *combined* total on 260-340, and 2,264 of these impossible values — 97.75% of them — fall in exactly that second band. That reading survives a second test: 1,675 of them also report a verbal score separately, and subtracting it leaves a mean of 165.26, back inside the legal range and within a point of the average of the values that were never out of range at all. A row reading `gre = 323` beside `gre_v = 161` is not a typo; it is an applicant answering a differently-worded question than the one the field label implies. The effect on the headline number is not marginal. Question 3 reports an average GRE Quantitative of 261.45, which no human being has ever scored; restricted to values the scale permits, the same column averages 165.74. Analytical Writing behaves the same way, inflated from 4.33 to 9.17 by placeholder entries of 99.99 on a six-point scale. This is the distinction the assignment asks me to draw, and it is sharper here than I expected: the database mathematically contains an average of 261.45, and that number is correctly computed, correctly rounded, and reproducible by anyone who runs the query — and it is still not a fact about GRE scores. It is a fact about a text box. The practical lesson is that a query can only be as meaningful as the agreement between what a field is called and what people put in it, and that on anonymous self-submitted data that agreement has to be tested rather than assumed. I have therefore reported the literal answers the questions ask for and attached the measurement of their distortion beside them, rather than quietly substituting cleaned figures and presenting them as though the underlying data were sound.